

HAOYUN ZHANG

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EDUCATION

University of Michigan, Ann Arbor, Michigan, USA Aug. 2025 – May. 2027(expected)

Master's Degree of Science major in Biomedical Engineering (BME),

- GPA: 4.0/4.0

Shanghai Jiao Tong University (SJTU), Shanghai, China Sept. 2021 – Jun. 2025

Bachelor's Degree of Engineering major in Biomedical Engineering (BME),
minor in Mathematics and Applied Mathematics

- A+/A Courses: Data Structure(96), Biomedical Signals and System(96), Medical Bioinformatics(94), Biomedical Image Processing(93), Stochastic Process(99), etc.
- GPA: 86.81/100; 3.65/4.3 (88.44; 3.76/4.3 Junior & Senior Year)

PUBLICATION¹

Journal:

- **Haoyun Zhang***, Wentao Zhang*, Shuai Zhao, Guangyu Xu, Yi Shen, Feng Jiang, An Qin[✉], Lei Cui[✉]. easySCF: A Tool for Enhancing Interoperability Between R and Python for Efficient Single-Cell Data Analysis. *Bioinformatics(Oxford, England)*, btae710. 25 Nov. 2024, doi:10.1093/bioinformatics/btae710
- Chen, Z.*, Lin, Z.*, Lin, Z., Zhang, Q., **Zhang, H.**, Li, H., Chang, Q., Sun, J., & Li, F.[✉] (2024). The applications of CT with artificial intelligence in the prognostic model of idiopathic pulmonary fibrosis. *Therapeutic advances in respiratory disease*, 18, 17534666241282538.
- Ruolin Mao*, **Haoyun Zhang***, Qing Chang*, et al. Progression and mortality of interstitial lung abnormalities in patients with lung cancer resection. (Under Review)
- Tiaoao Chen, ..., Lana X Gamire[✉]. HistToSpatialCNV: a deep learning method for predicting spatial copy number variations from histological images. (Under Review)
- Zeyu Chen*, **Haoyun Zhang***, et al. Artificial intelligence and multicenter multimodal data improves the evaluation of idiopathic pulmonary fibrosis prognosis. (Preparing for Submission)

Conference:

- Feng Li, Zeyu Chen, **Haoyun Zhang**, et al. The Prognosis of Idiopathic Pulmonary Fibrosis in Chinese Patients: A Multi-Center and Multi-Modal Cohort Study[J]. *American Journal of Respiratory and Critical Care Medicine*, 2025, 211: A1749-A1749. <https://doi.org/10.1164/ajrccm.2025.211.Abstracts.A1749>

RESEARCH EXPERIENCE

Query-Conditioned Patient Retrieval: Benchmark, Evaluation, and Method June. 2025 – Present

Supervisor: Prof. Joyce Yanran Wang @ University of Michigan-Ann Arbor

- *Reframing "similar patient" retrieval as a multi-relation problem: which patients are relevant depends on the clinical question being asked.*
- Benchmarking whether clinical retrieval systems return the right similar patients when relevance depends on the question asked, not a single global similarity metric.

Biology-Informed Multimodal Alignment of Single-Cell Foundation Models Jan. 2025 – Present

Supervisor: Prof. Jie Liu @ University of Michigan-Ann Arbor

- Designed FineCLS, a fine-grained branch decomposing each cell into 64 biology-defined modules (pathway/lineage gene sets paired with linked cCREs) via a zero-parameter prior router, yielding per-module interpretable embeddings alongside the single-vector global head.

¹*Both authors contributed equally to the work. ✉Corresponding Authors

- Outperforms the global head of the same model by +10% R@1 / +9% R@5 on cross-modal retrieval, +0.027 sub-celltype accuracy ($p < 0.001$) and +7.5% scIB integration score.
- Its module-agreement score serves as a calibrated confidence signal, beating any signal derivable from the global head by up to +0.14 AUC across four tissues and lifting R@1 4.5× (0.13 → 0.58) on the top-confidence decile.
- Drove the contrastive-objective and normalization design: replaced a degenerate log-of-sum multi-positive loss with a soft-AND SupCon ($\Sigma \log$) formulation and introduced dataset-level embedding centering for batch-effect removal — +42% in-distribution and +77% OOD retrieval R@1.

GNN-Based Inference of CNV Via Histopathology Images

Aug. 2024 – Dec. 2024

Supervisor: Prof. Lana Garmire @ University of Michigan-Ann Arbor

- Participation in projects of predicting chromosomal copy number variation (CNV) through H&E-stained histological images. A research article is being prepared for submission.
- Investigated and performed downstream analysis of CNV results, including phylogenetic trees, cell trajectory and pathway analysis.

Construction of a Prognostic Model for IPF based on Machine Learning

Apr. 2023 – Aug. 2025

Supervisor: Feng Li (Chief Physician) @ SHSMU² & **Jianqi Sun** (Research Professor) @ SJTU

- Constructed a prognostic model for rare disease idiopathic pulmonary fibrosis (IPF) based on machine learning algorithm by combining clinical indicators and imaging features
 - Enhanced deep imaging feature extraction by optimizing the Med3D (ResNet) model with attention mechanisms and segmentation-driven encoding
 - Improved survival prediction model accuracy, enhancing AUC by 11.3%, 6.1%, and 3.5% in 1-, 2-, and 3-year survival prediction, respectively, compared to the baseline (Cox proportional hazards model on clinical data)
- Provided technical expertise and maintained close collaboration with Shanghai Chest Hospital physicians and medical students throughout the entire workflow
 - Integrated clinical reasoning into ML models, focusing on identifying diffuse disease lesions and prioritizing key lung segmentation areas
 - Improved model interpretability and conducted subtyping and phenotyping of diseases, enhancing clinical comprehension and insight into pathological conditions

Development of Single-Cell Bioinformatics Packages

Oct. 2023 – Mar. 2024

Supervisor: Prof. An Qin @ SHSMU³

- Developed a unified data format to streamline data exchange in single-cell analysis workflows using Python and R, achieving a 30% improvement in computational efficiency while significantly reducing memory and disk usage.
- Enhanced the performance of existing GPT-based cell type annotation models by leveraging NCBI databases, improving accuracy and efficiency in genomic analysis.

INDUSTRIAL INTERNSHIP EXPERIENCE

Fosun | Algorithm Engineer – Deep Learning for Ultrasound

Feb. 2025 – Jun. 2025

- Developed deep learning models for pneumothorax diagnosis in lung ultrasound videos, optimized ROI localization and feature extraction pipeline, reducing computational cost by 60% and improving inference speed by 150%.
- Contributed to full-cycle medical AI product development, including model lightweighting (ONNX/TensorRT), preprocessing of DICOM/AVI formats, and deployment on edge devices.

²Shanghai Jiao Tong University School of Medicine, Shanghai Chest Hospital

³Shanghai Jiao Tong University School of Medicine, Shanghai Ninth People's Hospital

HONORS & AWARDS

- **National Golden Medal (TOP 0.01 %)**, China International College Students' Innovation Competition 2024 (The **Most Important** University Competition in China)
- **Honorable Mention(TOP 1.5 %)** , The 2nd World AI For Science Prize - Life Science Track: siRNA Drug Efficacy Prediction (Kaggle-like AI for Sciences Contest)
- **Honorable Mention**, 2022 Mathematical Contest In Modeling/Interdisciplinary Contest In Modeling
- Chen Yazhu (Academician, Chinese Academy of Engineering) Scholarships 2024 (**TOP 2%**)

SKILLS

- **Programming Languages:** Python(PyTorch, Tensorflow), R, Matlab, C++
- **Development & Deployment:** Docker, ONNX/TensorRT optimization, Linux development environment, Git/GitHub CI/CD
- **Other Tools:** SolidWorks, Latex, Unity, Arduino
- **Language:** English (Fluent, IELTS Total: 7.0); Chinese (Native Speaker)